MUGHEER ISLAM KHATTAK

Software Engineer | Backend & Full-Stack Systems

Abbottabad, Pakistan | Mugheer.ik@gmail.com | +92-320-5802402 | Linkedin | Github | Portfolio

PROFESSIONAL SUMMARY

Backend-focused Software Engineer with hands-on experience building authenticated APIs, database-backed applications, and full-stack systems through independent engineering projects. Skilled in Python, FastAPI, PostgreSQL, Redis, Docker, and automated testing, with practical experience in multi-tenant authorization, caching, rate limiting, database migrations, and CI/CD. Seeking a Software Engineering role focused on backend systems, reliability, and scalable application development.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Backend Engineering: FastAPI, REST API Design, SQLAlchemy, Pydantic, Alembic, JWT Authentication, RBAC, Multi-Tenant Authorization, Asynchronous Processing
Databases & Systems: PostgreSQL, Redis, Relational Data Modeling, Indexing, Migrations, Caching, Rate Limiting, Linux
Testing & Infrastructure: Pytest, HTTPX, Ruff, pre-commit, Docker, Docker Compose, Git, GitHub Actions, CI/CD
Frontend & Integration: React, Next.js, TypeScript, Gemini API Integration
Additional: Analytics engineering background: dbt, ETL/ELT, Power BI, DAX

EXPERIENCE

Independent Software Engineering Practice | *Self-Directed Projects* | *May 2025 – Present*

- Designed and built backend and full-stack systems using Python, FastAPI, PostgreSQL, SQLAlchemy, Redis, Next.js, and Docker.
- Built authenticated REST APIs with role-based access control, resource ownership, and multi-tenant data isolation.
- Applied automated testing, database migrations, containerization, and CI/CD across multiple complete projects.

SELECTED PROJECTS

LinkVault — URL Shortening & Analytics API

Python, FastAPI, PostgreSQL, Redis, Docker, GitHub Actions | github

- Built a URL-shortening REST API with authenticated link management, expiration handling, redirects, and click analytics.
- Implemented Redis caching with PostgreSQL fallback, reducing average redirect latency in local benchmarks; added Redis-backed rate limiting.
- Wrote automated tests for authentication, ownership, rate limiting, and cache-failure fallback; containerized with Docker and integrated CI.

TeamFlow — Multi-Tenant Task Management API

Python, FastAPI, PostgreSQL, SQLAlchemy, Docker, GitHub Actions | github

- Built a multi-tenant REST API for organizations, memberships, projects, and tasks using a modular monolith architecture.
- Implemented role-based access control (Owner/Admin/Member) with server-side enforcement and tenant-isolated data access.
- Designed the PostgreSQL schema with constraints and indexes; validated authorization with automated RBAC and cross-tenant access tests.

Doc Service — AI Document Processing Application

Python, FastAPI, Next.js, PostgreSQL, Gemini API, Docker Compose, GitHub Actions | github

- Built a full-stack document processing application supporting authenticated PDF, DOCX, and TXT uploads with user-level resource ownership.
- Implemented asynchronous text extraction and Gemini-powered analysis with processing status tracking and PostgreSQL result persistence.
- Containerized the application with Docker Compose and integrated database migrations, health checks, automated testing, and CI.

EDUCATION

Bachelor of Science in Software Engineering — COMSATS University Islamabad, Abbottabad Campus | 2021 – 2025 | CGPA: 2.96/4.00

Final Year Project (Team of 3): React Native/Firebase mobile news app with AI-assisted video summarization (Python/Flask).

CERTIFICATIONS

AWS Cloud Practitioner Essentials (AWS) · IBM DevOps & Software Engineering (IBM) · Google IT Automation with Python (Google) · Google Advanced Data Analytics (Google) · Microsoft Power BI Data Analyst (Microsoft)

Languages: English (Professional Working Proficiency) · Urdu (Native) · Pashto (Native/Professional)